



Reading: Print Services

IT professionals are often responsible for adding and updating printer drivers and settings. This may occur when a printer is added to a network, moved to a new location, or there is a software update. Along with updating drivers and settings on printers, IT may also be responsible for adding network printers to employee computers. Correct printer configuration saves time, supplies, and effort. This reading covers printing languages, basic printer configuration settings, printer sharing, printer security, and network scan services.

Printing languages

When choosing a print driver or troubleshooting issues with one, it is important to know which printing language the printer and computer operating system are using. Printing languages describe images on a screen to a printing device, so the printed output matches what is on screen. Printing languages are also called page description languages. Two of the most common printing languages are Printer Control Language and PostScript.

Printing languages can be either device-dependent or device-independent. Device-dependent means both the printer and computer are responsible for creating parts of the printed data. Device-independent means that the computer is solely responsible for creating the printed data. It is helpful for IT to know if the printing languages used are device-dependent or independent as it can help them troubleshoot whether printing errors are occurring because of the driver on the computer or the printer's hardware.

Printer Control Language (PCL)

Printer Control Language (PCL) is a printing language created by Hewlett-Packard that is used by many printer brands and computer operating systems. PCL is printing device-dependent because both the printer and computer are responsible for creating parts of the printed data. Because PCL is device-dependent, the output may not be the same on every printing device.

PostScript (PS)

PostScript was created by Adobe and is a printing language used by many printer brands but most commonly used in Macintosh systems. Unlike PCL, PostScript does not use the printer to create data. PostScript is device-independent, and the output is the same on any printer. If an error arises when PostScript is used, then it is usually an error with the driver on the computer.

Basic printer configuration settings

Configuration settings tell a printer how to complete a print job including the size, type of paper, number of sides, and use of color. IT professionals help employees change and select the correct

settings for their document. The following are basic configuration settings that can be adjusted using printer settings.

- **Orientation** is the direction in which a document is printed. The main options for most printers are portrait (vertical) and landscape (horizontal).
- **Print Quality** refers to the level of detail that both the paper and the print settings are set to. The higher the DPI (Dots Per Inch), the higher the resolution or quality of the print.
- **Tray settings** tell the printer which tray of paper to use for the print job. Different trays can hold different paper sizes and types. Telling the printer to select paper from the correct tray ensures that the document is printed as it was designed.
- **Duplex** allows for printing on both sides of the paper. Printers can print information on one side (simplex) or both sides of the paper. Many brochures, booklets, and packets are printed on both sides to save paper.

For more information on how to update printer settings for high-quality printing see the article in the reference section below.

Sharing a printer on a network

Printers can be shared on a network allowing multiple computers to access one printer across the network instead of having to be wired to the computer directly. IT professionals maintain and set up networks that include shared printers. For more information on sharing printers on your network read the article in the reference section below.

Network scan services

Network scan services allow a printer with scanning capabilities to create a file of a scanned image and upload or send it to a location on the network or in the cloud, or attach the file to an email and send it. Employees often need IT support for ways to use this type of technology. The following network scan services can be used for fast file uploads or attachments.

- **Email** scan service allows a document to be scanned directly from the printer to email.
- **Server Message Block (SMB) protocol** allows a document to be a shared resource once scanned by the printer.
- **Cloud services** enable a document to be scanned from the printer and uploaded directly to the cloud.

Printer security

Printer security protects access and tracks the activity of a print device. Printer security aims to ensure that only authorized users can use a printer. Setting up and monitoring proper security permissions falls under the job of an IT professional.

Some basic measures for limiting access to printers and tracking print activity are:

- **User authentication** commonly requires a user to enter a username and password before completing the print job.
- **Badges** are usually a physical card a user must scan at the printer to complete the print job.
- **Secured prints** require a user to enter a user-created code at the printer to complete the print job.
- **Audit logs** track users that have accessed the printer, including the date and time of use.

Key takeaways

IT support professionals are often responsible for printer management. It is helpful to know about printing languages, printer configuration, networking, and security.

- Printer Control Language is device-dependent, while Postscript is device-independent.
- Some basic printer configuration settings are orientation, print quality, tray settings, and duplex.
- Having a printer on a network enables multiple users to share printers.
- Network scan services allow a printer with scanning capabilities to create a file of a scanned image and upload or send it to a location on the network, on the cloud, or email.
- Printers have security and tracking features such as user authentication, badges, secured print, and audit logs.

Resources for more information

For more information about software and driver downloads for specific brand devices, review the links below.

[HP Customer Support - Software and Driver Downloads](#)

[Cannon Customer Support - Software and Driver Downloads](#)

[Xerox Customer Support - Software and Driver Downloads](#)

[Ricoh Customer Support - Software and Driver Downloads](#)

[HP - How to Update Printer Settings for the Highest Quality Printing](#)

[Microsoft Support - Share your network printer](#)

[Xerox - Scan a Document to an Email Address](#)

[HP Customer Support - Set up Scan to Network Folder](#)

[Dell - How to Configure Your PC or Server for SMB \(Server Message Block\) Scanning on Dell Laser Printers](#)

[Xerox - Scan to Cloud or Enable Remote Destination](#)
